

No.	Answer	Solution
4	B	$U = T + mg$ $T = U - mg$ $= \frac{2}{3}\pi r^3 \rho g - mg$ $= \frac{2}{3}\pi (0.500)^3 (1030)(9.81) - (200)(9.81)$ $= 683 \text{ N}$
5	D	Let tension in string and spring be T_1 and T_2 respectively